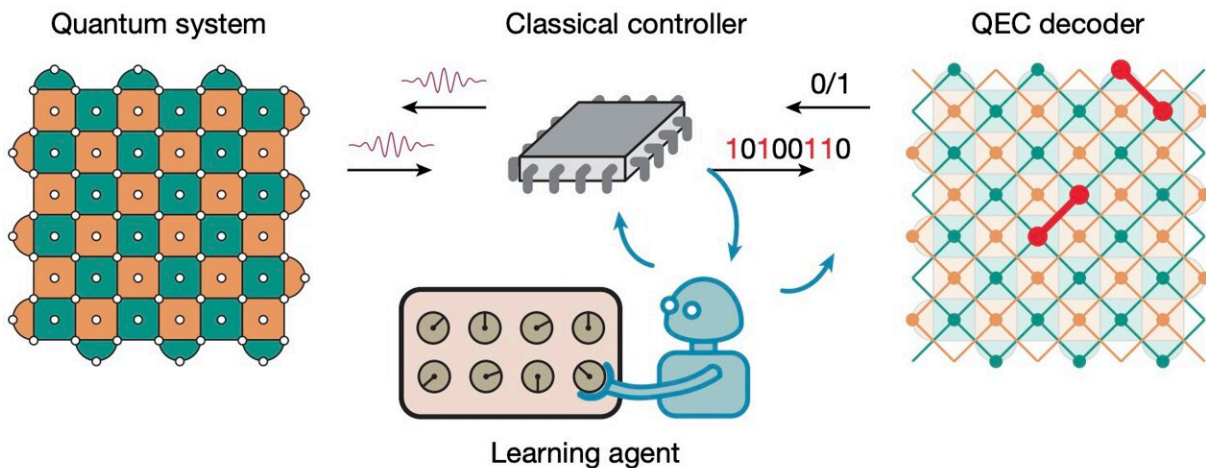


Faster quantum computers can learn from their own mistakes

July 15 2026, by Sam Jarman



Error detection events are re-used by the algorithm to help stabilize the quantum system during computation. Credit: *Nature* (2026). DOI: 10.1038/s41586-026-10759-2

Quantum computers promise to solve problems that would take even the fastest conventional supercomputers a vast amount of time, but the quantum information they store and process is extremely sensitive to even tiny disturbances from their surroundings. To keep these systems operating reliably, they need to be constantly recalibrated—interrupting their calculations in the process.

In a new experiment [published](#) in *Nature*, researchers led by Volodymyr

Sivak at Google Quantum AI developed a machine-learning approach that continuously adjusts a quantum computer as it works. Their approach could allow quantum calculations to run far longer without costly interruptions.

The recalibration problem

The building blocks of quantum information, quantum bits, or "qubits," are notoriously fragile. When subjected to even tiny changes in temperature, electrical currents or gradual drift in control electronics, the likelihood of errors can significantly increase.

To prevent this, modern quantum systems need to undergo regular calibration, in which the settings used to control each qubit are carefully adjusted to minimize mistakes. However, this requires calculations to stop completely while the machine is recalibrated—presenting a major obstacle for the long calculations that researchers hope to run on future quantum computers.

Reusing error data

To overcome this challenge, Sivak's team considered how quantum computers already monitor for errors as they run. Using specialized qubits, they can detect when something has gone wrong without disturbing the calculation itself.

Instead of using this information solely to identify errors, the team also fed it into a reinforcement learning algorithm. By making tiny adjustments to thousands of control settings and observing how the pattern of detected errors changed, the algorithm gradually learned which changes improved the system's stability. In turn, the quantum computer could learn from its own mistakes while continuing its

calculation.

Simulated boost in performance

The researchers tested this approach on Google's Willow superconducting quantum processor, deliberately introducing drift into the system to simulate subtle changes in the surrounding environment. With the learning algorithm continuously updating the control settings, the system became some 3.5 times more stable than existing error-correction methods. Crucially, this performance could be maintained even while the processor continued running.

Based on further simulations, Sivak's team suggested that their method could be scaled up to systems containing tens of thousands of adjustable control parameters without becoming significantly slower.

Preparing for bigger calculations

Although today's quantum computers aren't yet large enough for the recalibration problem to be a major limitation, the team's result addresses a challenge that will become increasingly important as the technology matures. By allowing quantum computers to continuously refine their own operation instead of pausing for recalibration, the technique could help future computers tackle much longer and more complex calculations.

Rather than treating errors as something to simply eliminate, the team's approach ultimately shows how errors can help quantum computers learn from their mistakes, becoming more stable over time.

More information: Volodymyr Sivak et al, Reinforcement learning control of quantum error correction, *Nature* (2026). [DOI:](#)

[10.1038/s41586-026-10759-2](https://phys.org/news/2026-07-faster-quantum.html)

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